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Laboratory Activity 3: GPIO and Button Control

I. Task

The task of this laboratory activity is to assemble a button and status LED circuit using an ESP32. Example 2 was tested first, and a second LED was then added to display the opposite state of the first LED.

II. GPIO Connections

The following GPIO connections were used in the circuit:

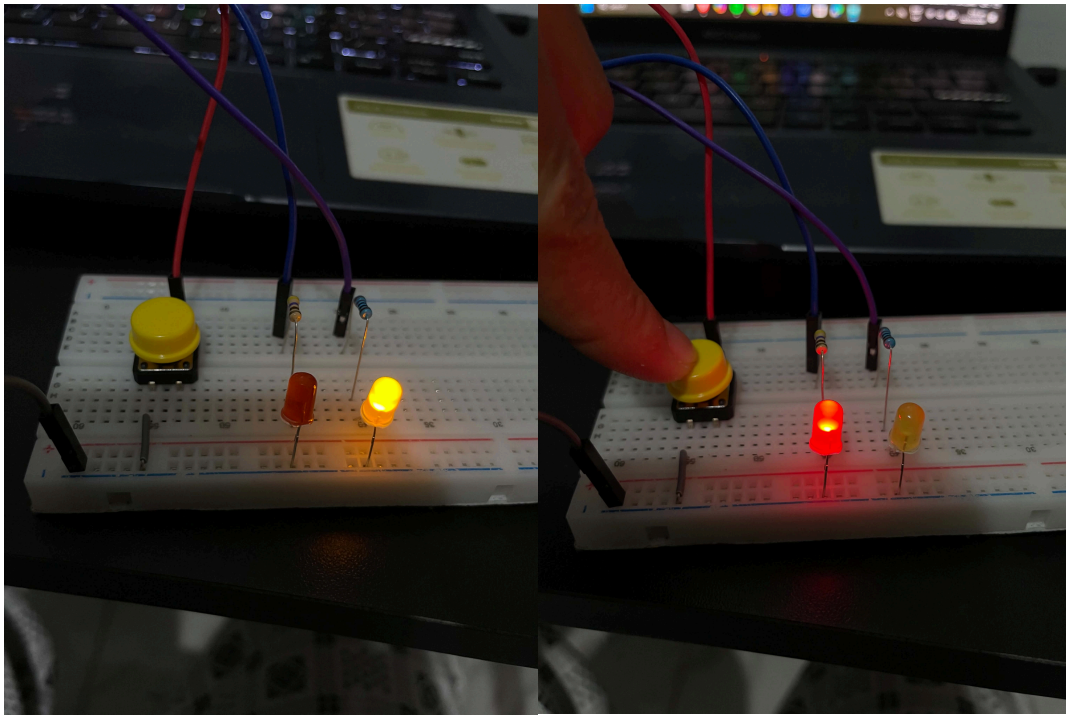
Component	ESP32 Connection
Push Button	GPIO 23 and GND
LED 1	GPIO18
LED 2	GPIO19
LED Ground	GND

Each LED was connected with a current-limiting resistor.

III. Observation

Button State	Input Reading	LED 1	LED 2	Observation
Released	HIGH	OFF	ON	LED 1 was OFF while LED 2 was ON.
Pressed	LOW	ON	OFF	LED 1 turned ON while LED 2 turned OFF.

Actual Photo Output



V. Explanation

When the push button was released, LED 1 was OFF while LED 2 was ON. When the button was pressed, LED 1 turned ON while LED 2 turned OFF. This showed that the two LEDs operated in opposite states as required. The push button was configured using INPUT_PULLUP. When the button was released, the input read HIGH because the internal pull-up resistor kept the GPIO pin at a HIGH level. When the button was pressed, the GPIO pin was connected to GND, causing the input to read LOW. After resetting the ESP32 while the button was released, LED 1 was initially OFF and LED 2 was ON. The button input also remained stable when released and did not randomly change its state.

VI. Conclusion

The laboratory activity was successfully completed using an ESP32, a push button, and two LEDs. The push button correctly produced HIGH when released and LOW when pressed because INPUT_PULLUP was used. The two LEDs also showed opposite states: when LED 1 was ON, LED 2 was OFF, and vice versa. The circuit remained stable when the button was released, confirming the expected GPIO input and output behavior.